

Session 9 Quiz - AI Hallucinations

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57029]

Complete the following definition: An AI Hallucination occurs when an AI makes information up but presents it as if it is _____.

- A. A joke
- B. True
- C. A question
- D. An opinion

Correct Answer: B. True

Explanation: Hallucinations are deceptive because the AI provides incorrect information with the same authority as factual data. The AI sounds confident even when it is making things up, which can trick users into believing false information.

Q2 [Type: MCQ | Difficulty: Easy | QID:57030]

Fill in the blank: AI models are often described as 'pattern _____' because they predict words without actually understanding their meaning.

- A. Engineers
- B. Detectives
- C. Parrots
- D. Architects

Correct Answer: C. Parrots

Explanation: This term highlights that AI repeats patterns found in data without comprehending the underlying concepts, just like a parrot repeats sounds without understanding meaning. Pattern parrots is a common analogy for large language models.

Q3 [Type: MCQ | Difficulty: Easy-Medium | QID:57031]

Complete the sentence: While accuracy is a matter of fact, confidence is simply a matter of _____ and presentation.

- A. Intelligence
- B. Tone
- C. Memory
- D. Logic

Correct Answer: B. Tone

Explanation: Confidence refers to how authoritative the AI sounds, which is a stylistic choice rather than a reflection of truth. The tone of voice (in text) can make an AI seem sure even when it is wrong.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57032]

Fill in the blank: When an AI lacks specific information, it often fills _____ with guesses that sound plausible based on patterns.

- A. Books
- B. Servers
- C. Internet
- D. Gaps

Correct Answer: D. Gaps

Explanation: AI models are designed to provide answers, so they use statistical guesses to bridge the holes (gaps) in their knowledge. They fill gaps with plausible sounding patterns rather than admitting they do not know.

Q5 [Type: MCQ | Difficulty: Easy-Medium | QID:57033]

Complete the following: One way to reduce hallucinations is to give the AI _____ to say 'I don't know' if it isn't sure.

- A. Money
- B. Hardware
- C. Permission
- D. Energy

Correct Answer: C. Permission

Explanation: Explicitly telling the AI it is okay to admit ignorance prevents it from feeling forced to make up an answer. By giving permission to say "I don't know," you reduce the chance of hallucinations.

Q6 [Type: True/False | Difficulty: Easy | QID:57034]

True or False: An AI can be 100% confident and 100% wrong at the same time.

- A. True
- B. False

Correct Answer: A. True

Explanation: AI confidence levels are based on word probability patterns, which can strongly point to an incorrect answer. Just like the movie night example where someone was sure the film would be good but was wrong, AI can be confidently wrong.

Q7 [Type: True/False | Difficulty: Easy-Medium | QID:57035]

True or False: If an AI provides a specific name of a study and a date (like 'The 2019 Harvard Study'), it is guaranteed to be a real source.

- A. True
- B. False

Correct Answer: B. False

Explanation: AI often generates fake names and dates that follow the pattern of real academic citations to seem more authoritative. The Brain Stretch activity with Dr. Sarah Johnson showed how AI can invent realistic but fake sources.

Q8 [Type: True/False | Difficulty: Easy-Medium | QID:57036]

True or False: NotebookLM is designed to hallucinate less because it focuses only on the sources you provide.

- A. False
- B. True

Correct Answer: B. True

Explanation: By restricting the AI's search area to specific documents, it is less likely to wander into guesswork from its general training data. This is like an open book test versus a test taken from memory.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57037]

True or False: Courts have ruled that companies, like airlines, are not responsible if their chatbot gives a customer the wrong information.

A. False

B. True

Correct Answer: A. False

Explanation: The Air Canada story showed that legal systems hold businesses accountable for the claims made by their automated systems. Courts have ruled that companies are responsible for their chatbots' misinformation.

Q10 [Type: True/False | Difficulty: Easy | QID:57038]

True or False: AI models hallucinate because they do not actually understand the facts they are reciting.

A. True

B. False

Correct Answer: A. True

Explanation: Because AI only predicts the next likely word based on patterns, it has no way to verify if those words correspond to reality. The Recap section described AI as a pattern parrot, which explains why hallucinations happen.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57039]

Which of the following describes the most dangerous state for an AI to be in for a user who needs facts?

- A. Confident and Right
- B. Confident and Wrong
- C. Unsure and Wrong
- D. Unsure and Right

Correct Answer: B. Confident and Wrong

Explanation: This is the hallucination zone where the AI sounds authoritative but provides false information, making it easy to trust blindly. Confident and wrong is much more dangerous than unsure and wrong because users are more likely to believe it.

Q12 [Type: MCQ | Difficulty: Medium | QID:57040]

In the 2023 New York lawyer case, what was the specific 'hallucination' that got the lawyer in trouble?

- A. The AI refused to answer questions about the law.
- B. The AI invented six court cases that did not exist.
- C. The AI gave the wrong date for a famous trial.
- D. The AI insulted the judge during the court hearing.

Correct Answer: B. The AI invented six court cases that did not exist.

Explanation: ChatGPT generated realistic looking legal citations and details for cases that had never actually occurred. The lawyer submitted these fake cases to the court, which led to serious consequences.

Q13 [Type: MCQ | Difficulty: Medium | QID:57041]

What kind of hallucination is happening if an AI tells you that 'Einstein invented the telephone'?

- A. Logical Hallucination
- B. Relationship Hallucination
- C. Source Hallucination
- D. Entity Hallucination

Correct Answer: B. Relationship Hallucination

Explanation: This type of error involves connecting two real things (Einstein and the telephone) in a way that is factually incorrect. The relationship between them is wrong, even though both entities are real.

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:57042]

Which of these is one of the '5 tricks' suggested to spot an AI hallucination?

- A. Only use AI late at night when fewer people are using it.
- B. Ask the AI to promise it is telling the truth.
- C. Assume that long answers are more accurate.
- D. Cross-check with reliable sources like Google.

Correct Answer: D. Cross-check with reliable sources like Google.

Explanation: Verifying AI claims against trusted, independent websites is the most effective way to catch factual errors. The other options are not reliable strategies.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:57043]

Why is 'Fills Gaps with Guesses' a common reason for hallucinations?

- A. The AI wants to play a game with the user.
- B. The AI has a very small memory and forgets things easily.
- C. AI is programmed to provide an answer even if it doesn't know the facts.
- D. AI models are trying to save battery power.

Correct Answer: C. AI is programmed to provide an answer even if it doesn't know the facts.

Explanation: Since the model's goal is to complete the text, it will use its training patterns to invent a likely sounding response rather than admitting it does not know. This is a core design feature that leads to hallucinations.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57044]

If you ask an AI to 'Tell me about dogs' versus 'Tell me 3 facts about Golden Retrievers you are sure about,' why is the second prompt better?

- A. Being specific limits the AI's guesswork.
- B. The second prompt is longer, and longer is always better.
- C. It makes the AI work faster.
- D. AI prefers specific breeds of animals.

Correct Answer: A. Being specific limits the AI's guesswork.

Explanation: Narrowing the scope and adding a requirement for certainty forces the model to focus on higher probability patterns. Specific prompts reduce the need for the AI to fill gaps with guesses.

Q17 [Type: MCQ | Difficulty: Medium | QID:57045]

What is an 'Entity Hallucination'?

- A. Giving the wrong date for the Fourth of July.
- B. Changing the color of a real object in a story.
- C. Inventing people, places, or things that do not exist.
- D. Failing to answer a math problem correctly.

Correct Answer: C. Inventing people, places, or things that do not exist.

Explanation: An entity hallucination involves the creation of entirely fictional entities like a fake professor, a made up city, or a nonexistent product. This is different from getting a real fact wrong.

Q18 [Type: MCQ | Difficulty: Medium | QID:57046]

How does providing context (like pasting an article) help reduce AI hallucinations?

- A. It allows the AI to connect to the live internet.
- B. The AI works from the provided text instead of guessing from its memory.
- C. It teaches the AI how to think like a human.
- D. It makes the AI's training data disappear.

Correct Answer: B. The AI works from the provided text instead of guessing from its memory.

Explanation: Grounding the AI in a specific source gives it a truth to refer to, which prevents it from wandering into random patterns. It is easier to summarize a page in front of you than to guess from memory.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57047]

Which tool would be best if you wanted to study a specific science article without the risk of the AI making up outside facts?

- A. NotebookLM
- B. ChatGPT
- C. Google Search
- D. Gemini

Correct Answer: A. NotebookLM

Explanation: NotebookLM is specifically designed to stick to provided sources, making it safer for research on specific documents. It has a lower risk of hallucinations compared to general purpose chatbots.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57048]

What is the key takeaway from the 'Pizza Hunt' and 'Weather Check' examples in the session?

- A. AI models are better at finding pizza than humans are.
- B. Pizza is a very difficult thing for AI to understand.
- C. Feeling confident does not mean you are actually right.
- D. You should never trust the weather forecast.

Correct Answer: C. Feeling confident does not mean you are actually right.

Explanation: These warm up examples show that humans, like AI, can feel 100% certain about something that turns out to be false. Confidence and truth are not the same thing.